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TECHNICAL REPORT OF
STRAY ANIMAL FOSTER & HEALTH TRACKING SYSTEM

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1. INTRODUCTION

1.1 Problem Statement

Stray animal management by local municipalities and NGOs often suffers from fragmented data systems. This disconnection leads to critical operational failures:

- **Overcrowded Fosters:** Volunteers receiving more animals than their safe capacity.
- **Missed Medical Dates:** Lost track of crucial vaccination and treatment schedules.
- **Unassigned Donations:** Inefficient tracking of supplies (food, medicine) donated to specific volunteers.
- **Adoption Bottlenecks:** Lack of centralized tracking for adoption requests and animal statuses.

1.2. The Solution:

This project designs a normalized, relational database system that acts as a central coordination layer. It seamlessly connects animals, volunteers, clinics, and donations to ensure data integrity, automate capacity checks, and streamline health tracking.

1.3. Project Objective and Justification

Animal welfare is a societal responsibility. This project aims to digitize the entire process of rescuing and adopting stray animals. The developed system provides a logistical tracking mechanism; it optimizes the capacity management of temporary shelters, systematizes vaccinations and veterinary visits, and ensures that donations reach the right targets. This tracking system minimizes operational errors and plays a life-saving role.

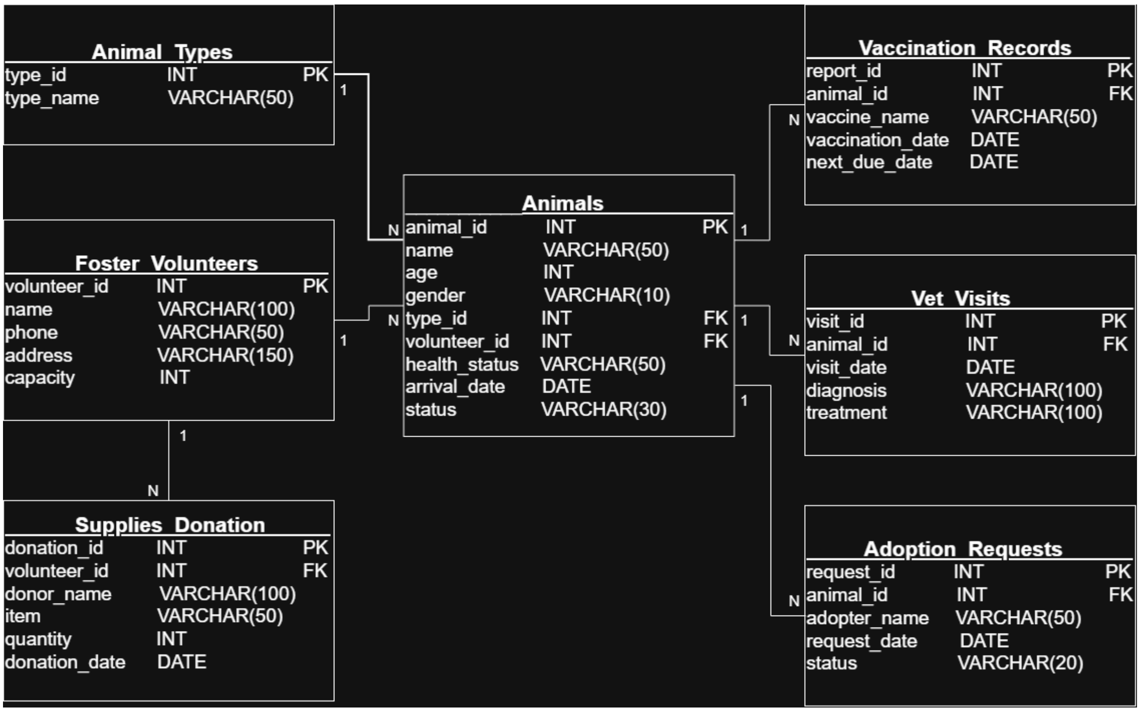
2. DATABASE DESIGN AND ARCHITECTURE

The system is built on a relational database model. Relationships between core entities are normalized up to Third Normal Form (3NF) to reduce redundancy and ensure data integrity. The schema is designed with 7 tables to accurately represent real-world operations and support efficient querying and reporting.

Table 2.1. Table Schemas

Table Name	Description	Key Relationships
Animal_Types	Master data for animal species/breeds.	1:N with Animals
Foster_Volunteers	Volunteer contact info and max safe capacity limit.	1:N with Animals & Supplies_Donations
Animals	Core identity, age, status, and current assignment.	FK to Foster_Volunteers & Animal_Types
Vaccination_Records	Logs of applied vaccines and upcoming due dates.	FK to Animals
Vet_Visits	Clinical diagnosis and treatment histories.	FK to Animals
Adoption_Requests	Tracking potential adopters and request statuses.	FK to Animals
Supplies_Donations	Tracking donated items assigned to specific volunteers.	FK to Foster_Volunteers

Figure 2.1. Entity Relationship Diagram (ERD)



3. TECHNICAL IMPLEMENTATION FEATURES

3.1. Business Rules Implemented in SQL

- **Capacity Integrity**: A volunteer's capacity must always be > 0 (CHECK constraint).

- **Age Validation:** An animal's age cannot be negative (CHECK >= 0).

- **Status Control:** Animal statuses are restricted to specific domains (e.g., 'Fostered', 'Adopted', 'Under Treatment').

- **Referential Integrity:** volunteer_id consistency ensures no animal is assigned to a non-existent volunteer.

3.2. Planned SQL Features

To analyze and manage this data effectively, we will utilize the following SQL techniques:

- **INNER / LEFT JOIN:** To link animals with their specific volunteers, types, and medical records.

- **GROUP BY + HAVING:** To monitor volunteer capacity utilization and donation distributions.

- **WITH (CTE):** To build structured analytical queries for upcoming vaccination deadlines
  .
- **VIEWS:** To create easily accessible virtual tables for daily operations (e.g., "Ready for Adoption" view).

- **ORDER BY & Aggregation:** Using COUNT(), MAX(), and date filtering for monthly reporting.

3.3. Sample Scenarios and SQL Queries

- **Scenario 1: The Active Roster (INNER JOIN)**

Lists all currently fostered animals along with their type and responsible volunteer's contact information.

```
SELECT
    a.name AS Animal_Name,
    t.type_name AS Species,
    a.age,
    f.name AS Foster_Name,
    f.phone
FROM Animals a
INNER JOIN Foster_Volunteers f
    ON a.volunteer_id = f.volunteer_id
INNER JOIN Animal_Types t
    ON a.type_id = t.type_id
WHERE a.status = ‘Fostered’
ORDER BY t.type_name, a.name;
```

- **Scenario 2: Capacity Control (GROUP BY + HAVING)**

Identifies volunteers who are currently at or over their maximum safe fostering capacity.

```
SELECT
    f.name AS Volunteer_Name,
    f.capacity AS Max_Capacity,
    COUNT(a.animal_id) AS Current_Fosters
FROM Foster_Volunteers f
LEFT JOIN Animals a
    ON f.volunteer_id = a.volunteer_id
GROUP BY f.name, f.capacity
HAVING COUNT (a.animal_id) >= f.capacity;
```

- **Scenario 3: Urgent Vaccination Pipeline (WITH RECURSIVE / CTE)**

Uses a Common Table Expression (CTE) to isolate upcoming vaccination needs due within the next 30 days.

```
WITH Upcoming_Vaccines AS (
    SELECT
        animal_id,
        vaccine_name,
        next_due_date
    FROM Vaccination_Records
    WHERE next_due_date BETWEEN CURRENT_DATE
        AND DATE(CURRENT_DATE, '+30 days')
)

SELECT
    a.name AS Animal_Name,
    f.name AS Volunteer_Name,
    u.vaccine_name,
    u.next_due_date
FROM Animals a
INNER JOIN Upcoming_Vaccines u
    ON a.animal_id = u.animal_id
INNER JOIN Foster_Volunteers f
    ON a.volunteer_id = f.volunteer_id
ORDER BY u.next_due_date ASC;
```

4. CONCLUSION

The designed Temporary Shelter and Health Monitoring System for Stray Animals aims to prevent confusion in the field by offering a database-driven solution. Thanks to the system's modular structure, future integration with mobile applications or automated notification systems (vaccination reminders via SMS/email) is possible. This infrastructure forms a strong foundation for sustainable animal welfare management.